

Design of a CMOS Broadband Transimpedance Amplifier With Floating Active Inductor

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Abstract—This work presents the design and performance of a transimpedance amplifier (TIA) implemented in a 180nm CMOS technology. The introduced TIA uses a floating active inductor (FAI) based on gyrator-C structure, which is why it can increase the bandwidth while occupying a smaller chip area. And we have explained the schematic and characteristics of the gyrator-C structure and FAI. In addition, the proposed TIA also uses the combination of capacitive degeneration, broadband matching network, and the regulated cascode (RGC) input stage, which turns the transimpedance amplifier (TIA) design into a fifth-order low-pass filter with butterworth response, to enhance the bandwidth and the gain. The TIA is implemented in 0.18 μm Rohm CMOS technology and a 1.8-V supply. The transimpedance amplifier achieves a transimpedance gain of 41 dB Ω and -3 dB frequency of 10 GHz with 0.1pF total input capacitance. Layout size of the proposed TIA is 180 μm $\times$ 118 μm .

Keywords—Floating active inductor (FAI), chip area, regulated cascode (RGC), transimpedance amplifier (TIA)

I. INTRODUCTION

Most of today's traffic is over the internet; therefore, data traffic has increased dramatically. Interest in high-speed optoelectronic devices and systems has increased again in recent years as the volume of telecommunications data has grown rapidly. Optical-fiber communication is the most promising system with high speed, low loss data and high bandwidth for handling this increase in the volume of information. In such a system, a receiver is the very important part.

In the receiver of optical communication system, the transimpedance amplifier (TIA) is the first electrical building block that converts the induced photo-diode (PD) current into a large voltage signal to be used in the digital processing unit. TIA is required to have high gain and wide bandwidth at the same time with low power dissipation [1]–[12].

It is well known that the inductive peaking effect is widely used to increase bandwidth and reduce parasitic capacitance effects. Inductor is one of the most important transimpedance amplifier (TIA) elements for high frequency application and RF systems. However, a common spiral inductor makes chips large and expensive because of its enormous size. Therefore, it is very important to reduce the inductor area required for TIA design. The active inductor (AI) is a gyrator that is implemented with a small number of transistors. Transistor-based active inductors (AI) were developed using two differential transconductors back-to-back connections in a negative feedback structure. The parasitic capacitance of the basic transistor is used as the load capacitance to realize the Gyrator-C topology.

Furthermore, a bandwidth enhancement method that based on the interaction of matching LC network, RGC input stage and capacitive degeneration stage, which turns the TIA into a fifth-order Butterworth low-pass filter can be used to enhance the bandwidth of the amplifier while providing efficient noise reduction. In this paper, we propose a new wide-band TIA using AI based on Gyrator-C topology, namely, Mahmoudi-Salama floating type, then, the design and postlayout simulation results of 10 GHz transimpedance amplifier (TIA) is presented using 180 nm CMOS technology.

This paper is organized as follows. Section II discuss the transimpedance amplifier based on traditional broadband design technique and illustrates a conventional TIA. Section III discuss the basic principle of gyrator-C networks, furthermore, a gyrator based differential floating active inductor (FAI) is introduced. Section IV describe the proposed TIA. The simulation results are shown in Section V and finally conclusions are summarized in Section VI.

II. CONVENTIONAL TIA BASED ON TRADITIONAL BROADBAND DESIGN TECHNIQUE

The conventional TIA is shown in Figure 1, where L1 and L2 are a spiral inductor, respectively. The circuit is composed of four parts, namely the matching network, the regulated cascode (RGC) input stage, the gain stage with capacitive degeneration and the source follower output stage [12].

The regulated cascode (RGC) stage presents a very small input impedance and therefore the lowest pole of the circuit is located within the TIA at RGC stage. In this TIA, we choose a small R_2 to avoid a possible peaking due to the zero generated by the local feedback of the RGC stage and a relatively large R_s to minimize its noise current contribution and signal loss.

The capacitive degeneration gain stage consists of M_3, R_3, R_b, C_b contributes a zero $(R_b C_b)^{-1}$ that is used to compensate the lowest pole determined at the RGC stage.

III. LOSSLESS FLOATING GYRATOR-C ACTIVE INDUCTORS

A. Principles of Gyrator-C active inductors

Inductor is one of the most important circuit elements for high frequency applications and RF systems. CMOS active inductors (AIs) have become a very hot research topic in recent years due to their many advantages over passive inductors. The AIs system can be implemented by a circuit

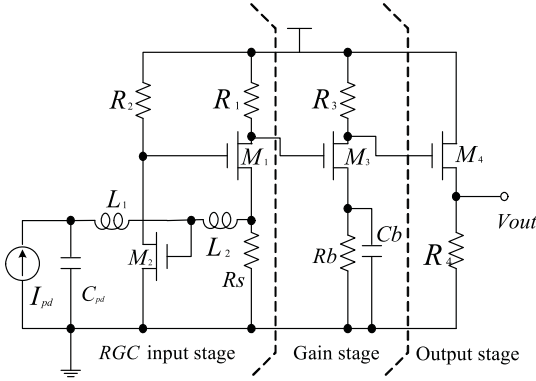


Fig. 1 Schematic of the conventional TIA.

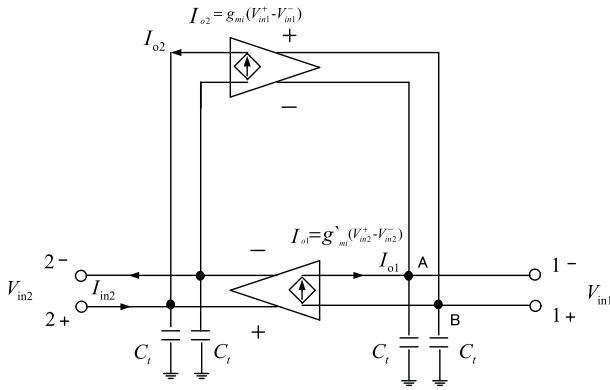


Fig. 2 Schematic of lossless floating gyrator-C active inductors.

that occupies a small chip area. In addition, AIs have large variable inductance and a high quality factor. On the other hand, AIs also have some disadvantages such as poor noise performance, and small dynamic range, nonlinear behaviour, dc power dissipation, sensitivity to process, compared with spiral inductors. Bandwidth extension topologies typically use on-chip inductors to compensate for capacitive effects, but the associated hardware costs are high. Therefore, it is very important to reduce the area of the inductor required for TIA design.

An active inductor is constructed with a small number of transistors for the implementation of the the gyrator. The gyrator consists of two back-to-back conncted transconductors [12]. When the port of gyrator is connected to a capacitor, as shown in Fig 2, the natwork is called the gyrator-C network. A gyrator-C network is said to be lossless when both the input and output impedances of the transconductor of the system are infinite, and the transconductances of the transconductors are constant.

For node A and node B, we have:

$$V_A = V_{in1}^- = -\frac{g_{mi}'}{sC_t}(V_{in2}^+ - V_{in2}^-), \quad (1)$$

$$V_B = V_{in1}^+ = \frac{g_{mi}}{sC_t}(V_{in2}^+ - V_{in2}^-), \quad (2)$$

$$I_{o2} = V_{in1}^+ = g_{mi}(V_{in1}^+ - V_{in1}^-). \quad (3)$$

Here, we substitute Equations (1) and (2) into Eq. (3), we have

$$I_{o2} = \frac{2g_{mi}g_{mi}'}{sC_t}(V_{in2}^+ - V_{in2}^-). \quad (4)$$

In this network, we can write

$$I_{in2} = -I_{o2}. \quad (5)$$

The admittance looking into port 2 of the gyrator-C network is given by

$$\begin{aligned} Y &= \frac{\frac{2g_{mi}g_{mi}'}{sC_t}(V_{in2}^+ - V_{in2}^-)}{V_{in2}^+ - V_{in2}^-} \\ &= \frac{2g_{mi}g_{mi}'}{sC_t} \\ &= \frac{1}{s\left(\frac{C_t}{2g_{mi}g_{mi}'}\right)}. \end{aligned} \quad (6)$$

Equation (6) reveals port 2 of the gyrator-C network behaves as a floating inductor with its inductance given by

$$L = \frac{\frac{C_t}{2}}{g_{mi}g_{mi}'}. \quad (7)$$

B. Mahmoudi and Salama active inductor

The floating active inductor proposed by Mahmoudi and Salama was used in the design of quadrature down converter for wireless application [13], [14]. The schematic of floating active inductor is shown in Fig. 3. It consists of a pair of differential transconductors and a pair of negative resistors at the output of the transconductors. M_8 and M_{16} are biased in the triode and behave as voltage-controlled resistors. They are added to the conventional cross-coupled configuration of negative resistors to provide the tunability of the resistance of the negative resistors without using a tail current source.

The differential active inductor is based on the conventional gyrator-C structure shown in Fig. 4. And according to the inductance relationship Eq. (7) when the gyrator-C network appears as a floating active inductor derived in the Sect. III-A, we can obtain

$$L = \frac{C}{g_{mi}g_{mi}'}. \quad (8)$$

IV. PROPOSED TIA

The conventional TIA is shown in Fig. 1, where L_1 and L_2 are a spiral inductor, respectively. In this work, we change L_2 from spiral inductor to floating active inductor, i.e., Mahmoudi-Salama floating active inductor, as shown in Fig. 5. As mentioned in the Section II, the proposed TIA is also composed of four parts: a matching network, the regulated cascode (RGC) input stage, the gain stage with capacitive degeneration and the source follower output stage.

Table I summarizes the device parameters of TIA.

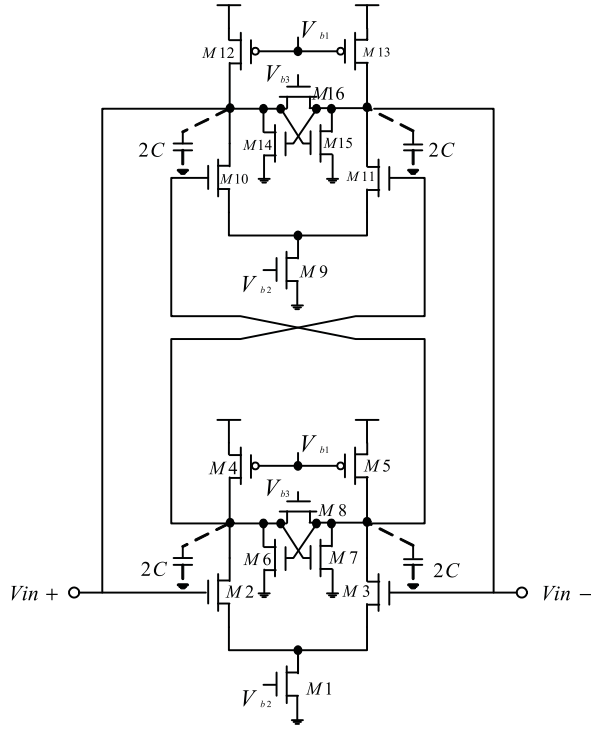


Fig. 3 Simplified schematic of Mahmoudi Salama floating active inductor.

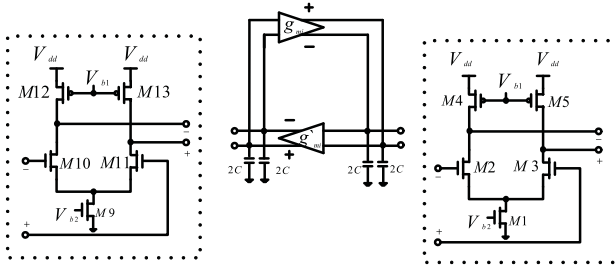


Fig. 4 Differential active inductor implementation based on the gyrator-C architecture.

V. POSTLAYOUT SIMULATION RESULTS

In order to evaluate the performance of the proposed TIA, the TIA is implemented in $0.18 \mu\text{m}$ Rohm CMOS technology. All post layout simulation results have been performed by using Cadence software tools.

The postlayout frequency response of the proposed TIA is shown in Fig. 6. The gain and the -3 dB bandwidth are $41 \text{ dB}\Omega$ and 10 GHz in the presence of a C_{pd} of 100 fF and an inductor L_1 of 3.2 nH at the input.

Figure 7 shows layout of the TIA. The TIA occupied area of the layout is $180 \mu\text{m} \times 118 \mu\text{m}$.

There are several advantages of our design. Firstly, as shown in Table II, our work provides reasonable gain and

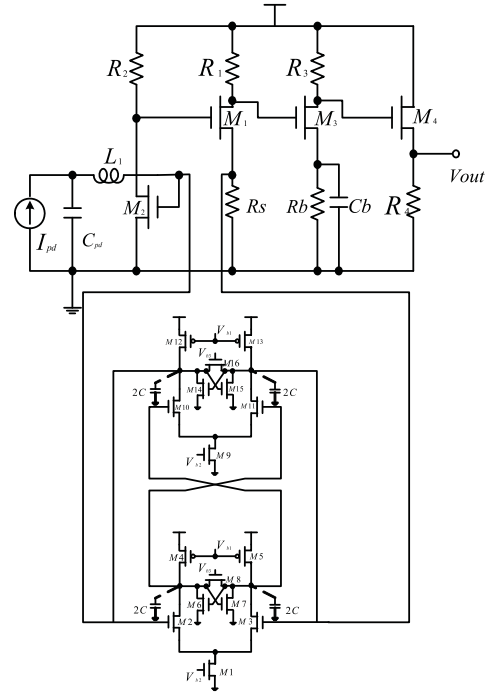


Fig. 5 Schematic of proposed TIA.

TABLE I Device Parameters of TIA circuit

Parameters	
$R_1 [\Omega]$	517.5
$R_2 [\Omega]$	345
$R_3 [\Omega]$	1.035k
$R_4 [\Omega]$	5.25k
$R_s [\Omega]$	517.5
$R_b [\Omega]$	70
$L_1 [\text{nH}]$	3.2
$C_{pd} [\text{pF}]$	0.1
$C_b [\text{pF}]$	0.7
$M_1 (L/W) [\mu\text{m}]$	0.18/3, m=10
$M_2 (L/W) [\mu\text{m}]$	0.18/2, m=20
$M_3 (L/W) [\mu\text{m}]$	0.18/2.5, m=2
$M_4 (L/W) [\mu\text{m}]$	0.18/2.5, m=1
AC Amplitude $[\mu\text{A}]$	100
DC $V_{dd} [\text{V}]$	1.8
Process	$0.18 \mu\text{m}$ standard CMOS

bandwidth for the low power consumption. Secondly, by using an active inductor consisting of Fig. (3) shown in the previous section rather than spiral inductor, it becomes possible to reduce the area and to expand gain and bandwidth.

VI. CONCLUSION

A 10 GHz bandwidth high performance transimpedance amplifier (TIA) with active inductor is presented. The TIA is implemented in $0.18 \mu\text{m}$ Rohm CMOS technology with 1.8 V supply voltage technology. Cadence tools is used in analyzing the performance of the TIA. The implemented TIA achieves over 10 GHz bandwidth and $41 \text{ dB}\Omega$ transimpedance gain in

TABLE II Comparison of characteristics of TIA

Reference	[2]	[20]	[7]	[11]	[21]	This work
Technology	180 nm	180 nm	180 nm	180 nm	80 nm	180 nm
Supply Voltage	1.8 V	1.8 V	1.8 V	1.8 V	1 V	1.8V
Transimpedance Gain (dB Ω)	53	55	51	54.3	52	41
Bandwidth (GHz)	8	7	30.5	7	20	10
C_{pd} (pF)	0.25	0.2	0.05	0.05	0.1	0.1
Power Dissipation (mW)	13.5	18.6	60.1	29	2.2	(11.18)
Chip Area	450 $\mu\text{m} \times 250 \mu\text{m}$	400 $\mu\text{m} \times 250 \mu\text{m}$	1170 $\mu\text{m} \times 460 \mu\text{m}$	230 $\mu\text{m} \times 45 \mu\text{m}$	140 $\mu\text{m} \times 70 \mu\text{m}$	180 $\mu\text{m} \times 118 \mu\text{m}$

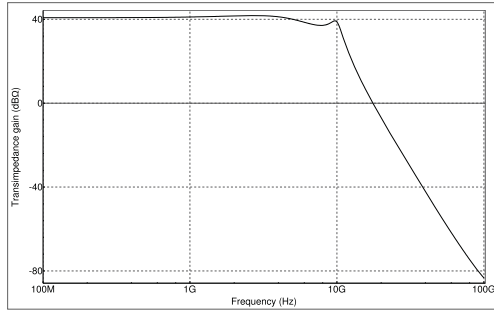


Fig. 6 Post-layout simulated frequency response of the proposed TIA.

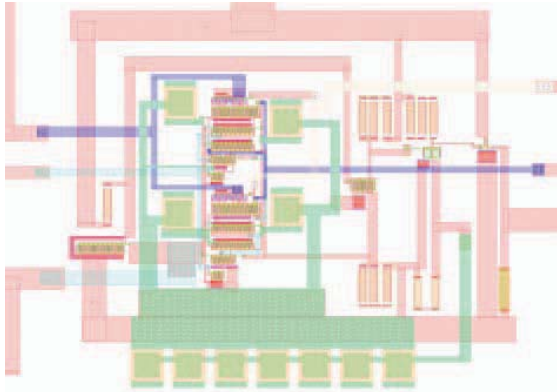


Fig. 7 Layout of the proposed TIA.

the presence of a 0.1 pF photodiode capacitance. By comparing other recently proposed TIAs, the proposed TIA is expected to take up a very small amount of chip space.

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REFERENCES

- [1] X. Chen, and Y. Takahashi, "Design and Analysis of A 10 GHz Transimpedance Amplifier Using Active Inductor in 0.18 μm CMOS process technology," *Proc. IEEE AVIC 2018*, pp.57–60, Oct. 31-Nov. 2, Chiang Mai, Thailand.
- [2] Z. Lu, K. S. Yeo, and J. G. Ma, "Broad-band design techniques for transimpedance amplifiers," *IEEE Trans. Circuit and Systems-I: Regular Papers*, vol. 54, no. 3, pp. 590–600, Mar. 2007.
- [3] B. Analui, and A. Hajimiri "Bandwidth enhancement for transimpedance amplifiers," *IEEE J. Solid-State Circuits*, vol. 39, no. 8, pp. 1263–1270, Aug. 2004.
- [4] Y. -H. Oh, and S. -G. Lee, "An inductance enhancement technique and its application to a shunt-peaked 2.5 Gb/s transimpedance amplifier design," *IEEE Trans. Circuit and Systems-II: Express Briefs.*, vol. 54, no. 3, pp. 624–628, Nov. 2004.
- [5] C. -H. Wu, C. -H. Lee, W. -S. Chen, S. -I. Liu, "CMOS wideband amplifiers using multiple inductive-series peaking technique," *IEEE J. Solid-State Circuits*, vol. 40, no. 2, pp. 548–552, Feb. 2005.
- [6] W. -Z. Chen, and C. -H. Lu, "Design and analysis of a 2.5-Gbps optical receiver analog front-end in a 0.35- μm digital CMOS technology," *IEEE Trans. Circuit and Systems-I: Regular Papers.*, vol. 53, no. 4, pp. 977–983, Apr. 2006.
- [7] G. -D. Jin, S. S. H. Hsu, "A 40-Gb/s transimpedance amplifier in 0.18- μm CMOS technology," *IEEE J. Solid-State Circuits*, vol. 43, no. 6, pp. 1449–1457, Jun. 2008.
- [8] H. S. Kimura *et al.*, "A 28 Gb/s 560 mW multi-Standard SerDes With single-stage analog front-end and 14-Tap decision feedback equalizer in 28 nm CMOS," *IEEE J. Solid-State Circuits*, vol. 49, no. 12, pp. 3091–3103, Dec. 2014.
- [9] Q. W. Song *et al.*, "Novel pre-equalization transimpedance amplifier for 10 Gb/s optical interconnects," *Journal of Semiconductors.*, vol. 36, no. 7, pp. 075002(1)–075002(5), Jul. 2015.
- [10] L. B. S. Samuel *et al.*, "An inductorless transimpedance amplifier design for 10 Gb/s optical communication using 0.18- μm CMOS," in *Proc. IEEE 2016 Int. Symp. Integrated Circuits (ISIC)*, pp. 1–4, Dec. 2016.
- [11] J. Abu-Taha, and M. Yazgi, "A 7 GHz compact transimpedance amplifier TIA in CMOS 0.18 μm technology," *Analog Integr. Circ. Sig. Process.*, vol. 86, pp. 429–438, 2016.
- [12] F. Yuan, *CMOS Active Inductors and Transformers*, Springer, New York, 2008.
- [13] F. Mahmoudi, and C. A. Salama, "8 GHz 1V, CMOS quadrature downconverter for wireless applications," *Analog Integr. Circ. Sig. Process.*, vol. 48, pp. 185–197, 2006.
- [14] F. Mahmoudi, "Quadrature Down-converter for Wireless Communications," Doctor thesis for the degree of Doctor of Philosophy, 2012.
- [15] X. Chen, and Y. Takahashi, "Evaluation of Wide-band Frequency Transimpedance Amplifier Using Active Inductors," *IEICE Technical Report*, vol. 117, no. 343, CAS2017-90, pp. 123–126, Dec. 2017.
- [16] X. Chen, and Y. Takahashi, "Small-signal Analysis of Trans-impedance Amplifier Using Mahmoudi-Salama's Floating Active Inductor," vol. ECT-18, pp. 53–56 (ETC-18-21), March. 2018.
- [17] B.-Y. Choi *et al.*, "A 1Gb/s Optical Transceiver Array Chipset for Automotive Wired Interconnects," in *Proc. IEEE Int. Symp. Circuits and Systems (ISCAS)*, pp. 181–184, May. 2007.
- [18] J. Han *et al.*, "A 2.5Gb/s ESD-Protected Dual-Channel Optical Transceiver Array," in *Proc. IEEE Asian Solid-State Circuits Conference.*, pp. 156–159, Korea, Nov. 2007.

- [19] H. L. Chen *et al.*, "Inductorless CMOS Receiver Front-End Circuits for 10-Gb/s Optical Communications," *Tamkang Journal of Science and Engineering*, vol. 12, no. 4, pp. 449–458, 2009.
- [20] Z. Lu *et al.*, "Design of a CMOS Broadband Transimpedance Amplifier With Active Feedback," *IEEE Trans. Very Large Scale Integration (VLSI) Systems*, vol. 18, no. 3, pp. 461–472, 2010.
- [21] C. -H. Wu, C. -H. Lee, W. -S. Chen, S. -I. Liu, "A Low-Power 20-GHz 52-dB Ω transimpedance amplifier in 80-nm CMOS," *IEEE J. Solid-State Circuits*, vol. 39, no. 6, pp. 885–894, Jun. 2004.
- [22] B.-Y. Choi *et al.*, "10Gbit/s 2mW inductorless transimpedance amplifier," in *Proc. IEEE Int. Symp. Circuits and Systems (ISCAS)*, pp. 1728–1731, 20–23 May. 2012.